

Reliable UDP Communication Using Stop-and-Wait in Mininet

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Timeout Used: 1.0 seconds

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Topology:

h1 ---- s1 ---- h2

h1 = Server

h2 = Client

Objective:

Implement reliable UDP communication using the Stop-and-Wait protocol in Mininet. Reliability is achieved using sequence numbers, acknowledgments (ACKs), timeout, and retransmission mechanisms.

Result Table

Paste the same table from your CSV:

Loss %	Timeout	Messages	Packets Sent	Retransmissions	Transfer Time	Status
0	1.0	10	10	0	1.3481	SUCCESS
5	1.0	10	11	1	2.353	SUCCESS
10	1.0	10	15	5	6.3888	SUCCESS

1. Why is ACK needed?

ACK is required to confirm that the receiver successfully received a packet. Without ACK, the sender cannot know whether the packet reached the destination.

2. Why is timeout needed?

Timeout allows the sender to detect packet loss. If an ACK is not received within the timeout period, the sender retransmits the packet.

3. Why can duplicate messages occur?

Duplicate messages occur when a packet is received successfully but its ACK is lost. The sender retransmits the packet, causing the receiver to receive the same message again.

4. What happens when packet loss increases?

As packet loss increases, retransmissions increase, transfer time becomes longer, and network efficiency decreases.

5. How is this method similar to TCP?

Both use acknowledgments, retransmissions, and timeout mechanisms to achieve reliable data delivery.

6. How is it different from TCP?

TCP provides built-in reliability, flow control, congestion control, and connection management. Stop-and-Wait UDP implements only basic reliability at the application layer.

Conclusion

The Reliable UDP Stop-and-Wait protocol was successfully implemented in Mininet. The experiments showed that increasing packet loss results in more retransmissions and higher transfer times while maintaining reliable delivery.